Serializability & Concurrency Control DBMS

> **Definition: Serializability** is the correctness criterion for concurrent transaction execution. A non-serial schedule is **serializable** if its execution outcome is equivalent to executing the transactions sequentially one after another (a Serial Schedule).

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1. Detailed Technical Explanation

1. Types of Schedules:

- **Serial Schedule:** Transactions execute strictly one after another (No interleaving). Guaranteed to be correct, but offers poor CPU utilization.

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2. Conflict Serializability:

Two operations **O_i** and **O_j** in a schedule are in **conflict** if and only if:

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Conflicting Operations Matrix:

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3. Precedence Graph (Serialization Graph) Test:

A directed graph **G = (V, E)** used to test Conflict Serializability:

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2. Core Concepts & Memory Keywords

- **Conflict Equivalent:** Two schedules can be transformed into one another by swapping non-conflicting operations.

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3. Must-Write Points for Exams

- A schedule is conflict serializable if it is conflict equivalent to some serial schedule.

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4. Quick Recall Flow

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